

PROJECT PLAN

AI Workforce OS

*An Enterprise Multi-Agent Platform for Business Operations***Course:** Web Services & Service-Oriented Architecture (IT644), Autumn 2026**Deliverable 2 (revised):** Project Plan — task timeline and milestones for the team and for individual members, team task assignments, tools, timeline chart and task board**Repository:** <https://github.com/BitecodesHub/hackemind>**Task board (Trello):** <https://trello.com/b/uVE4jHil/ai-workforce-os-wbs-effort-it644>**Team:** 8 members · **Committed effort:** 1716 person-hours against a 1600-hour benchmark · **Spread:** 212-216 h (1.9 per cent)

1. Response to the Project Review Feedback

Observation raised	How this plan responds
1. Ideas approved; novelty not uniform in every case	Section 2 states the differentiator: not a single assistant, but a governed multi-agent workforce with shared memory, cited enterprise retrieval, least-privilege tool execution, human approval gates and a complete audit trail.
2. Overall project scope was not fully quantifiable	Section 2 replaces prose scope with counted, named artifacts, split into committed and stretch scope.
3. Benchmark of 200 person-hours per member, 1600 in total, matched against task breakups and work assignment	Sections 4 to 9 cost every work package. Committed effort is 1716 hours; the lowest individual total is 212 hours, so every member clears the minimum from committed work alone.
4. Concerns to be addressed over the next two stages	This revised plan is the first stage. The Requirements Specification carries effort-costed user stories and acceptance criteria at the same granularity.
5. Integration and testing shown too late on the timeline, which is risky and not industry practice	Section 3 restructures this. Testing now begins in W4, not W15: continuous integration with test gates from the first pull request, per-service unit and contract tests inside every build phase, a deployed vertical slice in W6, an end-to-end skeleton in W8, and nightly integration runs from W9. The November phase is renamed System and Acceptance Testing because it consolidates and accepts, it does not begin testing. The test plan document is drafted in W4 and finalised in W13, ahead of its submission window.
6. Some members loaded more than others in the work distribution	Section 5 makes balance provable. Individual totals now fall in a 212-216 hour band, a spread of 4 hours or 1.9 per cent. No member is a coordinator only: the project lead now owns the integration test harness and the final report, and every member owns build work, test work and a documentation artifact.

2. Scope Quantification

Committed scope is what the 1716 hours buy. **Stretch scope** is stated above that total and is dropped first, so dropping it cannot take any member below the 200-hour benchmark. Reductions against the approved proposal are declared here rather than left implicit.

Artifact	Committed	Stretch (above the committed total)
Backend services	7 (gateway/auth, organization, orchestrator, shared memory, knowledge/RAG, MCP integration, analytics)	—
REST endpoints	48, documented in OpenAPI	—
AI employee agents	6 core: HR, Engineering Manager, Developer, Research, Marketing, Customer Support	4 extended: Analytics, Product Manager, Sales, Finance (proposal listed 8 extended; 4 deferred)
MCP server integrations	6: Gmail, Google Calendar, Slack, GitHub, Jira, Google Drive	4: Instagram, LinkedIn, CRM, AWS/Kubernetes
Knowledge connectors	6: Drive, Notion, Confluence, GitHub wiki, PDF, DOCX	—
UI screens	14 named screens	8 further screens
Data stores	3: PostgreSQL, Redis, Qdrant	Neo4j knowledge graph
Database tables / event topics	18 tables, 6 topics	—
Automated tests	approximately 150, written alongside the code they cover	—
Containers and manifests	8 images with Kubernetes manifests	—

Stretch scope is costed at 188 hours. Full ambition is therefore 1904 hours; the plan commits to 1716.

3. Testing and Integration Strategy

Testing is a thread running the length of the project, not a phase near the end. The table below states when each kind of testing starts and who owns it. Every build work package in Section 9 includes its own test effort, so no phase can be declared complete with untested code.

Activity	Starts	Cadence and owner	Exit condition
Continuous integration with test gates	W4 (17 Aug)	Every pull request; Afif	No merge to main unless tests pass
Unit and contract tests per service	W4, with the first code	Service owner, in the same card	Coverage threshold from the test plan
Deployed vertical slice on staging	W6 (6 Sep)	Afif with Param	One authenticated request served end to end
Cross-service integration tests	W8 (20 Sep)	Aum owns the harness; owners add cases	Gateway to orchestrator to agent to knowledge base
Nightly automated integration run	W9 (21 Sep)	Nightly in CI; Afif triages	A red night is fixed before new feature work
Per-module evaluation suites: agents and retrieval	W9, with the first agent and connector	Panthil (agents), Fahim (retrieval)	Each agent and connector evaluated before the next begins
MCP sandbox integration tests	W6, with the first server	Ismail, per server	Each server green on its sandbox before the next
User interface and accessibility tests	W6	Yash, per screen	Component tests per screen; journeys from W11
Test plan document	Drafted W4, finalised W13	Afif	Submitted already exercised for nine weeks
System and acceptance testing (P8)	W15 (2 Nov)	Whole team, Afif coordinating	Consolidation and acceptance, not first execution

4. Estimating Basis

Hours are built up from per-artifact rates, not apportioned from the benchmark. Each rate includes the tests for that artifact, which is why build phases carry more hours than in the earlier draft.

Unit of work	Rate	Unit of work	Rate
REST endpoint, with unit and contract tests	2 h	Knowledge connector, with tests	9 h
UI screen, with component tests	7 h	Service scaffold with CI and test harness	12 h
MCP server, with OAuth and sandbox tests	20 h	Integration test suite per handoff	8 h
AI agent: persona, tools, evaluation suite	18 h	Design or specification document	10-24 h

5. Effort Model and Workload Balance

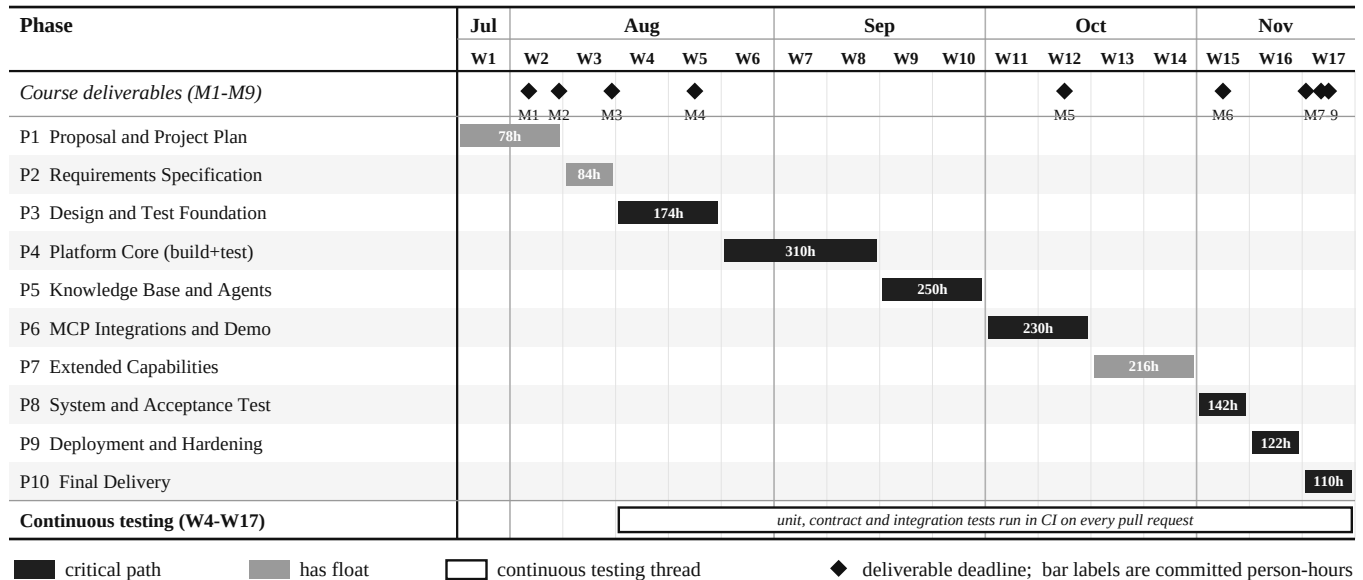
Rows are members, columns are the ten phases, cells are committed person-hours including the test effort for that work. Totals are the natural sum of the work packages in Section 9.

Member	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	Total	vs 200
Aum Parmar	16	12	24	26	24	26	24	20	20	20	212	+12
Love Shah	8	10	22	40	34	28	30	16	14	12	214	+14
Panthil Shah	8	10	22	28	36	36	30	18	12	14	214	+14
Fahim Saiyad	8	10	22	38	36	26	28	18	14	14	214	+14
Ismail Mansuri	8	10	22	44	36	30	24	18	12	12	216	+16
Yash Doshi	8	12	22	34	32	30	28	16	16	16	214	+14
Param Patel	8	10	18	58	28	26	26	16	16	10	216	+16
Afif Momin	14	10	22	42	24	28	26	20	18	12	216	+16
Phase total	78	84	174	310	250	230	216	142	122	110	1716	+116
Intensity, h/week/person	4.9	10.5	10.9	12.9	15.6	14.4	13.5	17.8	15.2	13.8	12.6	

Balance. Individual totals run from 212 to 216 hours, a spread of 4 hours (1.9 per cent). Every member exceeds the 200-hour benchmark from committed work alone, and every member carries build work, test work and a named documentation artifact, so no role is oversight only. The team commits 1716 hours against 1600, a margin of 116 hours which is the contingency reserve. Average load is 12.6 hours per member per week; the highest single member-week is 20 hours.

6. Timeline Chart

Ten phases across the seventeen course weeks (W1 begins Monday 27 July 2026; W17 has four working days). Bar labels are committed person-hours. The outlined band is the continuous testing thread of Section 3, which runs from W4 to the end. Diamonds mark the deadlines in Section 7.



Phase schedule, effort and ownership

Phase	Weeks	Dates	Hours	Cards	Lead	Deputy
P1 Proposal and Project Plan	W1-2	27 Jul-9 Aug	78	13	Aum	Afif
P2 Requirements Specification	W3	10-16 Aug	84	14	Aum	Param
P3 System Design and Test Foundation	W4-5	17-30 Aug	174	29	Aum	Yash
P4 Platform Core (build and test)	W6-8	31 Aug-20 Sep	310	52	Param	Love
P5 Knowledge Base and First Agents	W9-10	21 Sep-4 Oct	250	42	Fahim	Panthil
P6 MCP Integrations and Mid-Semester Demo	W11-12	5-18 Oct	230	38	Ismail	Yash
P7 Extended Capabilities and Hardening	W13-14	19 Oct-1 Nov	216	36	Love	Panthil
P8 System and Acceptance Testing	W15	2-8 Nov	142	24	Afif	Param
P9 Deployment and Hardening	W16	9-15 Nov	122	20	Afif	Ismail
P10 Final Delivery	W17	16-19 Nov	110	18	Aum	Yash

7. Milestones

#	Date	Course deliverable (team milestone)
M1	Wed 5 Aug	Project Proposal Document
M2	Sun 9 Aug	Project Plan: timeline chart and task board
M3	Sun 16 Aug	Requirements Specification (BRD), effort-costed
M4	Thu 27 Aug	System Design Document; OpenAPI v1.0 frozen; test plan v0.1
M5	Week of 12 Oct	Mid-Semester Demonstration
M6	Week of 2 Nov	Test Plan and Test Cases (already exercised since W4)
M7	Mon 16 Nov	Final Application on a public URL (code-complete Sun 15 Nov)
M8	Wed 18 Nov	Final Project Report; presentation, demo and poster from 18 Nov
M9	Thu 19 Nov	Peer Review and Contribution Report

Internal integration checkpoints (answering observation 5)

#	Date	Checkpoint and exit condition	Owner
I1	Sat 22 Aug	CI pipeline green on main with unit-test gate enforced	Afif
I2	Sun 6 Sep	Vertical slice deployed to staging: one authenticated request end to end	Afif, Param
I3	Sun 20 Sep	End-to-end skeleton: gateway to orchestrator to agent to knowledge base	Aum
I4	Sun 4 Oct	Nightly integration run green for five consecutive nights	Afif
I5	Fri 16 Oct	Full demo path integrated and rehearsed	Aum, Ismail
I6	Sun 1 Nov	Feature freeze; only defect fixes and hardening after this date	Aum
I7	Sun 8 Nov	System and acceptance testing complete; bug log burnt down	Afif

Individual milestones, each with a verifiable exit condition

Member	Individual milestones
Aum Parmar	OpenAPI v1.0 frozen — 27 Aug · integration test harness running — 20 Sep · demo dry run passed — 16 Oct · final report assembled — 18 Nov
Love Shah	Orchestrator routes a two-agent task with tests — 20 Sep · memory persists context across sessions — 4 Oct · asynchronous workflow with load test — 1 Nov
Panthil Shah	HR and Research agents with evaluation suites — 4 Oct · six core agents demo-ready — 16 Oct · full regression green — 8 Nov
Fahim Saiyad	Ingestion indexes a Drive folder with tests — 20 Sep · cited retrieval with accuracy report — 4 Oct · seeded corpus accepted — 8 Nov
Ismail Mansuri	Gmail, Calendar, Slack servers pass sandbox tests — 20 Sep · GitHub, Jira, Drive with approval hooks — 4 Oct · permission-scope suite green — 8 Nov
Yash Doshi	Auth and organization UI on staging with component tests — 20 Sep · chat workspace usable — 4 Oct · approvals queue demo-ready — 16 Oct · poster — 15 Nov
Param Patel	Auth service consumable with contract tests — 13 Sep · approval-queue API live — 11 Oct · security review signed — 15 Nov
Afif Momin	CI green with test gate — 22 Aug · staging environment complete — 20 Sep · nightly runs green — 4 Oct · public URL live — 15 Nov

8. Team Task Assignments

Each phase has one accountable lead and one deputy, so every member holds a leadership slot and every module has a named second who can take it over (Section 12). Contributor hours per phase are the columns of Section 5; the named work is in Section 9.

Member	Student ID	Module owned	Deputy for	Documentation artifact owned
Aum Parmar	202512004	Project Lead and System Architecture	—	Architecture and design document; final report
Love Shah	202512058	Agent Orchestration and Shared Memory	Aum Parmar (architecture)	Orchestration sequence diagrams
Panthil Shah	202512050	AI Employee Agents	Fahim Saiyad (knowledge base)	Agent catalogue and prompt documentation
Fahim Saiyad	202512103	Knowledge Base and RAG Pipeline	Panthil Shah (agents)	Knowledge base operations runbook
Ismail Mansuri	202512075	MCP Integration Layer	Yash Doshi (frontend)	MCP permission scope matrix
Yash Doshi	202512094	Frontend (Next.js Dashboard)	Ismail Mansuri (integrations)	UI style guide; poster and video
Param Patel	202512005	Backend Platform: Gateway, Auth and Organizations	Afif Momin (operations)	API documentation and Postman collection
Afif Momin	202512063	DevOps, Continuous Testing and Documentation	Param Patel (backend)	Test plan; user manual; deployment runbook

9. Individual Task Timelines

Each work package decomposes on the Trello board into cards of four to eight hours, approximately 286 in total. Test effort is inside each build package, not deferred.

Aum Parmar (202512004) — Project Lead and System Architecture — 212 h

Phase	Weeks	Work package (test effort included)	Hours	Due
P1	W1-2	Proposal, project plan, effort model and Trello board design	16	9 Aug
P2	W3	BRD consolidation, use-case diagrams, acceptance criteria	12	16 Aug
P3	W4-5	Service boundaries, OpenAPI v1.0, architecture diagram, design document	24	27 Aug
P4	W6-8	Contract reviews; integration test harness scaffold; vertical-slice integration	26	20 Sep
P5	W9-10	End-to-end skeleton integration (I3); cross-service integration tests; risk log	24	4 Oct
P6	W11-12	Mid-semester demo scenario, script, dry run and presentation	26	18 Oct
P7	W13-14	Integration oversight; feature-freeze control (I6); final report drafting	24	1 Nov
P8	W15	System test coordination, defect triage, acceptance sign-off	20	8 Nov
P9	W16	Release coordination and security review sign-off	20	15 Nov
P10	W17	Final report assembly, presentation, peer-review compilation	20	19 Nov

Love Shah (202512058) — Agent Orchestration and Shared Memory — 214 h

Phase	Weeks	Work package (test effort included)	Hours	Due
P1	W1-2	Orchestration scope input; weekly progress reports	8	9 Aug
P2	W3	Orchestration user stories and acceptance criteria	10	16 Aug
P3	W4-5	Orchestrator design, shared-memory schema, sequence diagrams	22	27 Aug
P4	W6-8	Orchestrator skeleton, routing, agent registry, with unit and contract tests	40	20 Sep
P5	W9-10	Shared memory on PostgreSQL and Redis; handoffs; integration tests	34	4 Oct
P6	W11-12	Demo workflow orchestration, approval-gate integration, web integration	28	18 Oct
P7	W13-14	Event-driven asynchronous workflows, retries, dead-letter handling, load tests	30	1 Nov
P8	W15	System testing of orchestration paths; defect fixes	16	8 Nov
P9	W16	Orchestrator hardening and deployment support	14	15 Nov
P10	W17	Final report sections, demo support, peer review	12	19 Nov

Panthil Shah (202512050) — AI Employee Agents — 214 h

Phase	Weeks	Work package (test effort included)	Hours	Due
P1	W1-2	Agent scope input; weekly progress reports	8	9 Aug
P2	W3	Agent roles, capabilities and acceptance criteria	10	16 Aug
P3	W4-5	Agent personas, prompt architecture, evaluation plan	22	27 Aug
P4	W6-8	Agent base framework, tool-calling scaffold, agent test harness	28	20 Sep
P5	W9-10	HR and Research agents end to end, each with its evaluation suite	36	4 Oct
P6	W11-12	Developer, Engineering Manager, Marketing and Support agents, with regression tests	36	18 Oct
P7	W13-14	Agent quality benchmarking, prompt tuning, evaluation harness extension	30	1 Nov
P8	W15	Agent acceptance testing; defect fixes	18	8 Nov
P9	W16	Agent catalogue and prompt documentation; deployment support	12	15 Nov
P10	W17	Final report sections, demo, peer review	14	19 Nov

Fahim Saiyad (202512103) — Knowledge Base and RAG Pipeline — 214 h

Phase	Weeks	Work package (test effort included)	Hours	Due
P1	W1-2	Knowledge base scope input; weekly progress reports	8	9 Aug
P2	W3	Knowledge base requirements and acceptance criteria	10	16 Aug
P3	W4-5	Ingestion pipeline design, chunking and embedding strategy, schema	22	27 Aug
P4	W6-8	Ingestion pipeline, Qdrant store, Drive and PDF connectors, with connector tests	38	20 Sep
P5	W9-10	Cited retrieval; Notion, Confluence, wiki, DOCX connectors; accuracy tests	36	4 Oct
P6	W11-12	Retrieval tuning and relevance evaluation for the demonstration	26	18 Oct
P7	W13-14	Accuracy harness, re-ranking, chunking optimization	28	1 Nov
P8	W15	Seeded corpus; system retrieval acceptance tests	18	8 Nov
P9	W16	Deployment, index migration, operations runbook	14	15 Nov
P10	W17	Final report sections, demo, peer review	14	19 Nov

Ismail Mansuri (202512075) — MCP Integration Layer — 216 h

Phase	Weeks	Work package (test effort included)	Hours	Due
P1	W1-2	Integration scope input; weekly progress reports	8	9 Aug
P2	W3	Integration requirements and permission model	10	16 Aug
P3	W4-5	MCP permission scope matrix, OAuth flows, approval-hook design	22	27 Aug
P4	W6-8	Gmail, Calendar and Slack MCP servers, each with sandbox integration tests	44	20 Sep
P5	W9-10	GitHub, Jira and Drive MCP servers, with integration tests	36	4 Oct
P6	W11-12	Scoped permission enforcement, approval hooks, audit logging, demo integration	30	18 Oct
P7	W13-14	Credential rotation, rate-limit handling, permission-scope test suite	24	1 Nov
P8	W15	System integration testing across all servers; defect fixes	18	8 Nov
P9	W16	Secret management and integration hardening	12	15 Nov
P10	W17	Final report sections, demo, peer review	12	19 Nov

Yash Doshi (202512094) — Frontend (Next.js Dashboard) — 214 h

Phase	Weeks	Work package (test effort included)	Hours	Due
P1	W1-2	UI scope input; weekly progress reports	8	9 Aug
P2	W3	Initial wireframes and interface requirements	12	16 Aug
P3	W4-5	Near-final Figma mockups, design system, component inventory	22	27 Aug
P4	W6-8	App shell, authentication and organization UI (4 screens), with component tests	34	20 Sep
P5	W9-10	Chat workspace; knowledge search UI with citations (3 screens), with tests	32	4 Oct
P6	W11-12	Approvals queue, agent configuration, dashboards (4 screens); end-to-end journeys	30	18 Oct
P7	W13-14	Analytics dashboard, notification centre, audit-log viewer (3 screens)	28	1 Nov
P8	W15	End-to-end UI, accessibility and responsive testing; defect fixes	16	8 Nov
P9	W16	UI polish, error states; project poster design	16	15 Nov
P10	W17	Project video walkthrough; final report sections, peer review	16	19 Nov

Param Patel (202512005) — Backend Platform: Gateway, Auth and Organizations — 216 h

Phase	Weeks	Work package (test effort included)	Hours	Due
P1	W1-2	Backend scope input; weekly progress reports	8	9 Aug
P2	W3	Authentication and organization requirements	10	16 Aug
P3	W4-5	ER schema, authentication contracts, RBAC model	18	27 Aug
P4	W6-8	Gateway, auth service (JWT/OAuth2), organization and user services, RBAC, with tests	58	20 Sep
P5	W9-10	Vertical slice and orchestrator integration; audit-log service; integration tests	28	4 Oct
P6	W11-12	Approval queue APIs; notification service, with tests	26	18 Oct
P7	W13-14	Analytics APIs, extended endpoints, Postman collection, with tests	26	1 Nov
P8	W15	Backend system and acceptance testing; defect fixes	16	8 Nov
P9	W16	Security review, hardening, rate limiting, API documentation	16	15 Nov
P10	W17	Final report sections, demo, peer review	10	19 Nov

Afif Momin (202512063) — DevOps, Continuous Testing and Documentation — 216 h

Phase	Weeks	Work package (test effort included)	Hours	Due
P1	W1-2	Git repository, branch protection, Trello boards, AI prompt log	14	9 Aug
P2	W3	Non-functional requirements and the test approach	10	16 Aug
P3	W4-5	CI pipeline with unit-test gate (I1), Docker baselines, test plan v0.1	22	27 Aug
P4	W6-8	Dev and staging environments; vertical slice deployed (I2); CI runs tests on every pull request	42	20 Sep
P5	W9-10	Nightly integration run (I4); coverage reporting; build optimization	24	4 Oct
P6	W11-12	Monitoring, tracing and the demonstration environment	28	18 Oct
P7	W13-14	Autoscaling, log aggregation, test plan finalisation, user manual	26	1 Nov
P8	W15	System test execution coordination, bug log burn-down (I7)	20	8 Nov
P9	W16	Deployment to the target cluster, public URL, operations runbook	18	15 Nov
P10	W17	Peer-review collection, submission packaging, final report sections	12	19 Nov

10. Repository, Tools and Task Board

Repository. <https://github.com/BitecodesHub/hackemind> — a single monorepo holding the seven services, the web client, the infrastructure manifests and, under **docs/**, every course deliverable. The brief requires each deliverable in the Git repository as well as Google Classroom, so the repository is a submission channel and not only a code store. **main** is protected: no direct pushes, one approving review from the module deputy named in Section 8, and continuous integration must pass. Branches are named for module and card, for example **mcp/gmail-server**.

Planning tool. Trello, named in the brief; this supersedes the proposal's Jira, which remains only an MCP integration target inside the product. Two boards sit in one free workspace: **Board A, WBS and Effort**, ten lists one per phase, every card carrying owner,

due date, phase label and hours — the evidence board and source of the submitted screenshot — and **Board B, Sprint Board** on a two-week cadence, where Blocked is a red label rather than a list so a card keeps its workflow position. Titles carry the hours because a title survives the JSON export and is legible in a screenshot: **[P4][Backend] Gateway routing — Param — est 16h**, with / **act 14h** on completion.

Board A as built. Live at <https://trello.com/b/uVE4jHil/ai-workforce-os-wbs-effort-it644>: ten phase lists holding all eighty work packages, each card carrying its owner and committed hours in the title. Each package is decomposed into four-to-eight-hour cards as its phase opens, reaching roughly {sum(CARDS)} cards over the semester. Visibility is opened to the instructor and teaching assistants before submission, so they read it without consuming free-tier seats. Board B opens with Sprint 1 in W3.

Figure: Board B, the sprint board, as it opens for Sprint 1 — Sprint 0 closed, the requirements cards loaded, later phases held as work packages in the backlog.

Backlog P4-P10 work packages	P4 Platform Core (build+test) 52 cards / 310 h	P5 Knowledge Base & Agents 42 cards / 250 h	P6 MCP Integrations & Demo 38 cards / 230 h
	P7 Extended & Hardening 36 cards / 216 h	P8 System & Acceptance Test 24 cards / 142 h	P9 Deployment 20 cards / 122 h
	P10 Final Delivery 18 cards / 110 h		
Sprint Backlog S1 - 14 cards / 84 h	[P2][Backend] Auth + org requirements - Param - est 10h due 16 Aug	[P2][Orchestr] User stories - Love - est 10h due 16 Aug	[P2][Agents] Roles + acceptance criteria - Panthil - est 10h due 16 Aug
	[P2][RAG] KB requirements - Fahim - est 10h due 16 Aug	[P2][MCP] Integration + permission model - Ismail - est 10h due 16 Aug	[P2][Frontend] Wireframes - Yash - est 12h due 16 Aug
	[P2][QA] Non-functional + test approach - Afif - est 10h due 16 Aug	[P2][Docs] BRD consolidation - Aum - est 12h due 16 Aug	
In Progress 0 cards	no cards		
In Review 0 cards	no cards		
Done S0 - 13 cards / 78 h	[P1][Docs] Project proposal submitted - Aum - est 10h / act 10h 5 Aug	[P1][Docs] Project plan + effort model - Aum - est 6h / act 6h 9 Aug	[P1][DevOps] Git repo + branch protection - Afif - est 6h / act 6h 9 Aug
	[P1][DevOps] Trello boards + prompt log - Afif - est 8h / act 8h 9 Aug		

Task board screenshot: Board A as built

The screenshot displays the Trello board 'AI Workforce OS - WBS & Effort (IT644)'. The board is organized into six columns, each representing a project phase. Each column contains several cards, each representing a task. The cards include details such as the task name, owner, and estimated/actual hours. The columns are: P1 Proposal and Project Plan, P2 Requirements Specification, P3 System Design and Test Foundation, P4 Platform Core (build and test), P5 Knowledge Base and First Agents, and P6 MCP Integrations & Demo. The interface includes a top navigation bar with icons for inbox, planner, board, and switch boards, and a bottom bar with a 'Share' button.

Figure 1: live Trello board, captured part 1 of 2 — <https://trello.com/b/uVE4jHil/ai-workforce-os-wbs-effort-it644>



Figure 2: live Trello board, captured part 2 of 2 — <https://trello.com/b/uVE4jHil/ai-workforce-os-wbs-effort-it644>

Definition of done. A code card is done when its pull request is merged with unit and contract tests passing in continuous integration, the OpenAPI specification updated, the change deployed to staging and the module deputy's review recorded. A document card is done when the phase lead has reviewed it and it is committed to the repository. No card is done on the strength of code alone.

11. Dependencies and Critical Path

The critical path runs from the design contracts through the platform core, the knowledge base and agents, the integrations and demonstration, then system testing, deployment and delivery. P7 carries float because part of its content is stretch scope. A slip on any handoff below moves the final delivery.

Handoff	From	To	Date
OpenAPI v1.0 frozen	Aum	All service owners	Thu 27 Aug
CI green with test gate (I1)	Afif	All members	Sat 22 Aug
Vertical slice on staging (I2)	Afif, Param	All members	Sun 6 Sep
Authentication service consumable	Param	Yash, Ismail	Sun 13 Sep
Orchestrator API stable	Love	Panthil	Sun 20 Sep
End-to-end skeleton (I3)	Aum	All members	Sun 20 Sep
Retrieval API with citations	Fahim	Panthil, Yash	Sun 4 Oct
Demo path integrated (I5)	Panthil, Ismail, Param	Aum	Fri 16 Oct
Feature freeze (I6)	All members	Afif	Sun 1 Nov
All modules code-complete	All members	Afif (deployment)	Sun 15 Nov

12. Effort Tracking, Reallocation and Evidence

- **Recording.** Estimated hours are fixed at sprint planning; actual hours are updated at the Friday review, and every logged block references a commit, a pull request, a document revision or a minuted meeting, so hours are corroborated rather than asserted.
- **Roll-up.** A committed script parses the weekly Trello export into a planned-versus-actual table per member, stored under docs/effort/, with a burn-up chart against the 1716-hour commitment in the final report.
- **Balance monitoring.** The same weekly roll-up reports the spread between the highest and lowest cumulative actual hours. A spread above 15 per cent is corrected at the next sprint planning by moving backlog cards, so the balance in Section 5 is maintained in fact and not only in plan.
- **Cadence and reallocation.** Two-week sprints on phase boundaries; planning Monday, two short stand-ups, review and retrospective Friday, which also produces the weekly progress report. A member more than 15 per cent behind hands work to the named deputy. Two consecutive sprints behind, or an absence beyond one sprint, is raised with the instructor. Released stretch hours move to named committed work, never to reduce a member below 200 hours.

13. Risks, Assumptions and Contingency

Risk	Owner	Likelihood	Trigger	Response
Free cloud tier cannot host Kubernetes, an event bus and three data stores	Afif	High	Credits over 50 per cent consumed by W12	Decided in advance: k3s on a single virtual machine, Redis Streams in place of Kafka for the demonstration, Kubernetes manifests demonstrated locally
Member unavailable or falls behind	Aum	Medium	Two missed stand-ups, or 15 per cent variance	Named deputy assumes the module; stretch scope dropped first
Workload drifts out of balance	Aum	Medium	Weekly roll-up shows spread above 15 per cent	Backlog cards moved at the next planning session
Critical-path handoff slips	Aum	Medium	A Section 11 date missed by more than three days	Deputy pulled in; stretch dropped; contingency margin applied
Integration debt accumulates	Afif	Medium	Nightly integration run red for two nights	Feature work stops until the run is green
Social-platform OAuth approval delayed	Ismail	High	No approval by W11	Those servers are stretch; the demonstration uses the six committed servers
Language-model API spend exceeds budget	Panthil	Medium	80 per cent of the monthly cap	Smaller model for routine calls, cached responses, capped retries

Assumptions. Each member sustains about 12.6 hours a week for seventeen weeks alongside other courses. The institute calendar, including examinations and holidays inside W1 to W17, is overlaid at the first Friday review and the plan re-baselined if a week is lost. Language-model API access sits within a stated monthly budget, and all eight members remain on the team.

14. Submission Package

Committed to <https://github.com/BitecodesHub/hackemind> under docs/deliverable-2/ and uploaded to Google Classroom:

- This plan, with the timeline chart (Section 6) and the task board (Section 10); the chart also exported separately from GanttProject as PDF and PNG
- A screenshot of the live Trello Board A, together with its URL (<https://trello.com/b/uVE4jHil/ai-workforce-os-wbs-effort-it644>)
- The Trello import CSV of every work package, and the AI prompt log required by the brief